

NETWORK ANALYSIS

- DIRECTED GRAPH
- UNDIRECTED GRAPH
- ADJACENCY MATRIX
- EDGE LIST
- ADJACENCY LIST
- PLANAR GRAPH
- GEODESIC GRAPH
- GRAPH ECCENTRICITY
- GRAPH DIAMETER
- EULERIAN PATH
- HAMILTONIAN PATH

- DEGREE CENTRALITY
- DEGREE CENTRALITY UNDIRECTED GRAPH
- DEGREE DISTRIBUTION
- DENSITY
- BETWEENNESS
- CLOSENESS
- CONNECTED COMPONENT
- STRONGLY CONNECTED COMPONENT
- WEAKLY CONNECTED COMPONENT
- GIGANT COMPONENT
- STRONG AND WEAK TIES
- CLUSTERING (GLOBAL AND LOCAL)
- STRUCTURAL HOLES

- RANDOM GRAPHS
 - ERDOS-RENYI
 - WATTS-STROGATZ MODE
 - BARABASI-ALBERT MODEL
 - BARABASI-BIANCONI MODEL

- SCALE-FREE PROPERTIES
- RANDOM WALK MODEL
- THE WEB GRAPH
- SEARCH ENGINES
 - WEB CRAWLERS
 - BFS ARLGRITHM
 - RANK
 - TF-IDF
 - COSINE SIMILARITY

- MATH BACKGROUND FOR PageRank
 - MARKOV PROCESS

- GOOGLE PageRank
- TOPIC-SENSITIVE PageRank
- WEB SPAM
- CO-CITATION & BIBLIOGRAPHIC COUPLING
 - CO-REFERENCE

- HUBS AND AUTHORITIES
 - HITS

- PATTERNS OF RELATIONS
 - STRUCTURAL EQUIVALENCE
 - AUTOMORPHIC EQUIVALENCE
 - SIMILARITY MEASURES

JACCARD SIMILARITY
COSINE SIMILARITY
PEARSON CORRELATION COEFFICIENT

ASSORTATIVITY
HOMOPHILY
ASSORTATIVE AND DISASSORTATIVE MIXING
ASSORTATIVE AND DISASSORTATIVE NETWORKS

COMMUNITIES IN NETWORKS
ZACHARY KARATE CLUB
GRAPH BISECTION
RECURSIVE BISECTION
MULTI-LEVEL GRAPH PARTITIONING
MUTUALITY OF TIES: MAXIMUM CLIQUES
FREQUENCY OF TIES: STRONG AND WEAK COMMUNITIES
FREQUENCY OF TIES: K-CORES
CORE DECOMPOSITION
CLOSENESS OR REACHABILITY: K-DISTANCE CLIQUES
COMMUNITY DETECTION
LABEL PROPAGATION
THE GIRVAN-NEWMAN ALGORITHM
MODULARITY VS ASSORTATIVITY
LOUVAIN METHOD
LEIDEN ALGORITHM

INFOMAP
OVERLAPPING COMMUNITIES
SPEAKER-LISTENER LABEL PROPAGATION ALGORITHM

NETWORK DIFFUSION: EPIDEMIC SPREADING
SIS EPIDEMIC MODEL
SIR EPIDEMIC MODEL
SIRS EPIDEMIC MODEL
SIRS EPIDEMIC MODEL + SMALL WORLD

DYNAMICS: SOCIAL CONTAGION
SIMPLE CONTAGION
COMPLEX CONTAGION
THRESHOLD MODEL

NETWORK ROBUSTNESS
PERCOLATION THEORY
INVERSE PERCOLATION THEORY

ROBUSTNESS OF SCALE-FREE NETWORKS
MOLLOY-REED CRITERION
BARABASI ET AL. EXPERIMENT

CASCADING FAILURES
FAILURE PROPAGATION MODEL
BRANCHING MODEL
MOTTER AND LAI MODEL